

ML Scan Parameters

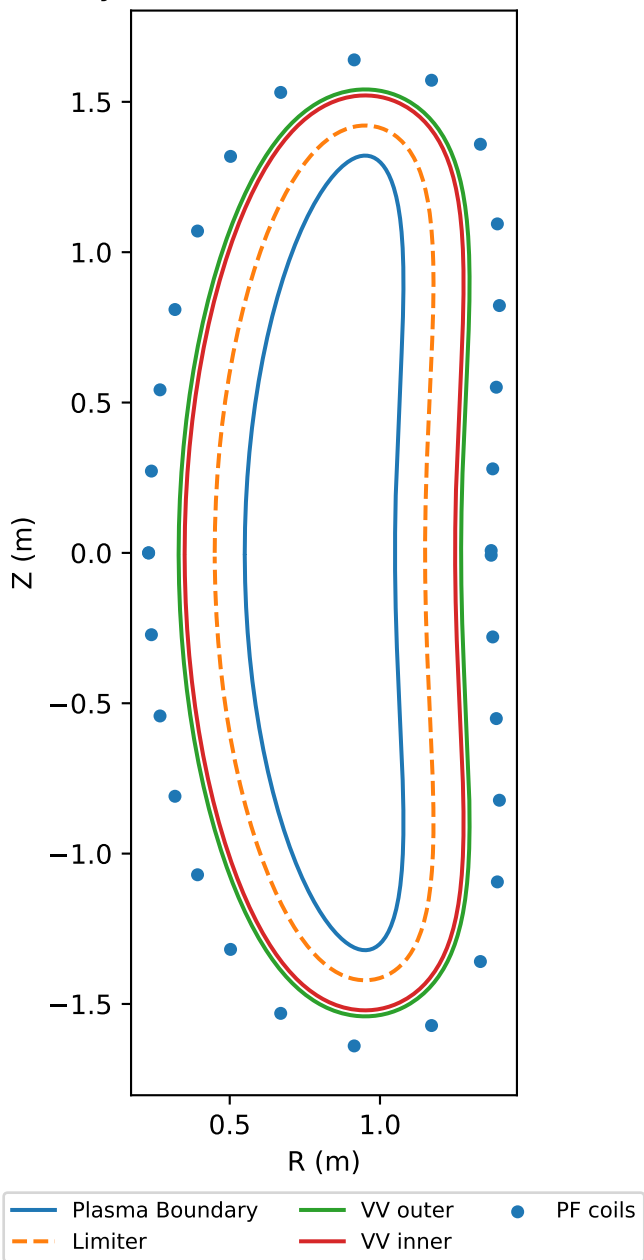
C-shape plasma $(r,z) = (A + B\sin t + C\cos 2t, F + G\cos t - H\sin 2t)$

A=0.9000 B=0.2500 C=0.1000
G=1.3000 H=0.1200 F=0.0
R0=0.8000 m kappa_target=5.200 delta=-0.120
B0=1.8 T Ip=1.00 MA

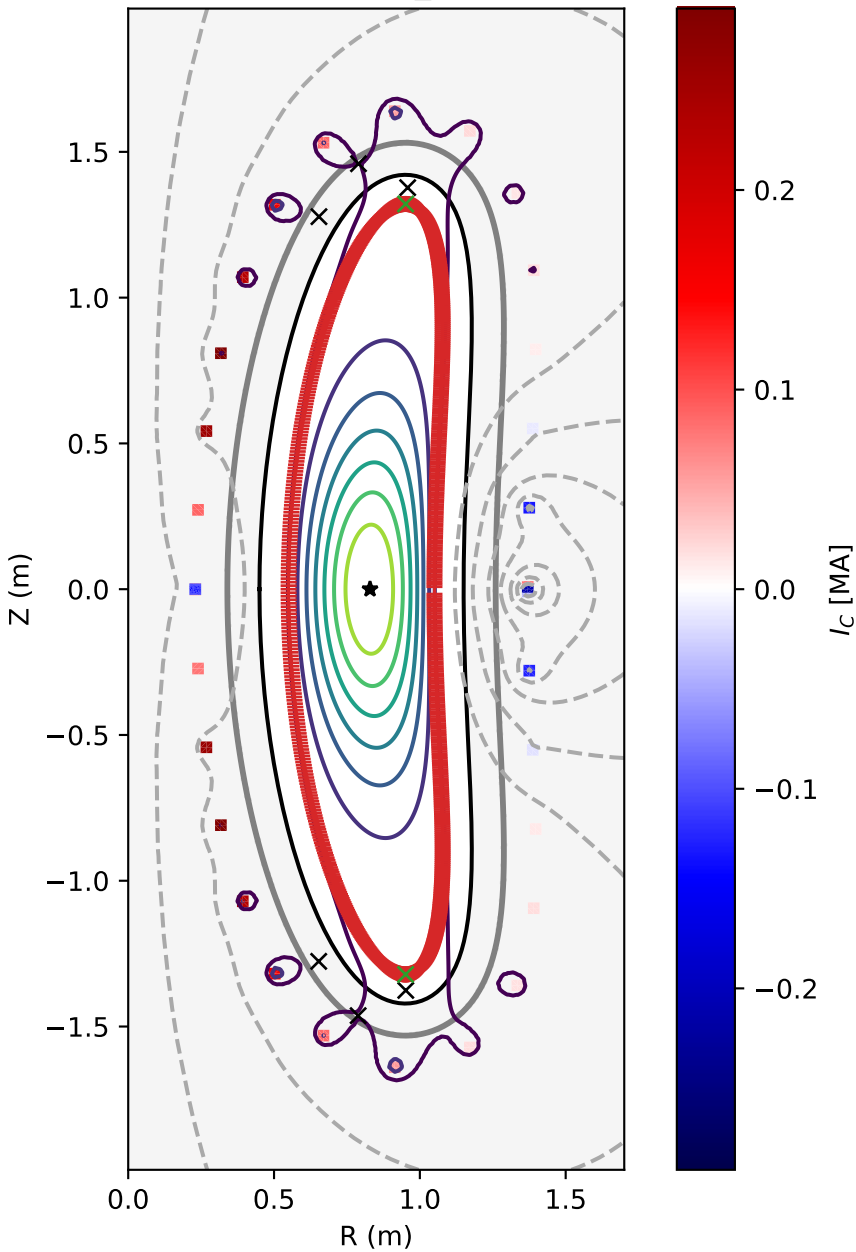
Plasma profiles:
pp_prof: $(1 - \psi^{2.033})^{0.935}$
ffp_prof: $(1 - \psi^{0.938})^{1.800}$

Scan params: B=2.5 C=10 G=13 H=12
alpha_pp=2.033 gamma_pp=0.935
alpha_ff=0.938 gamma_ff=1.800

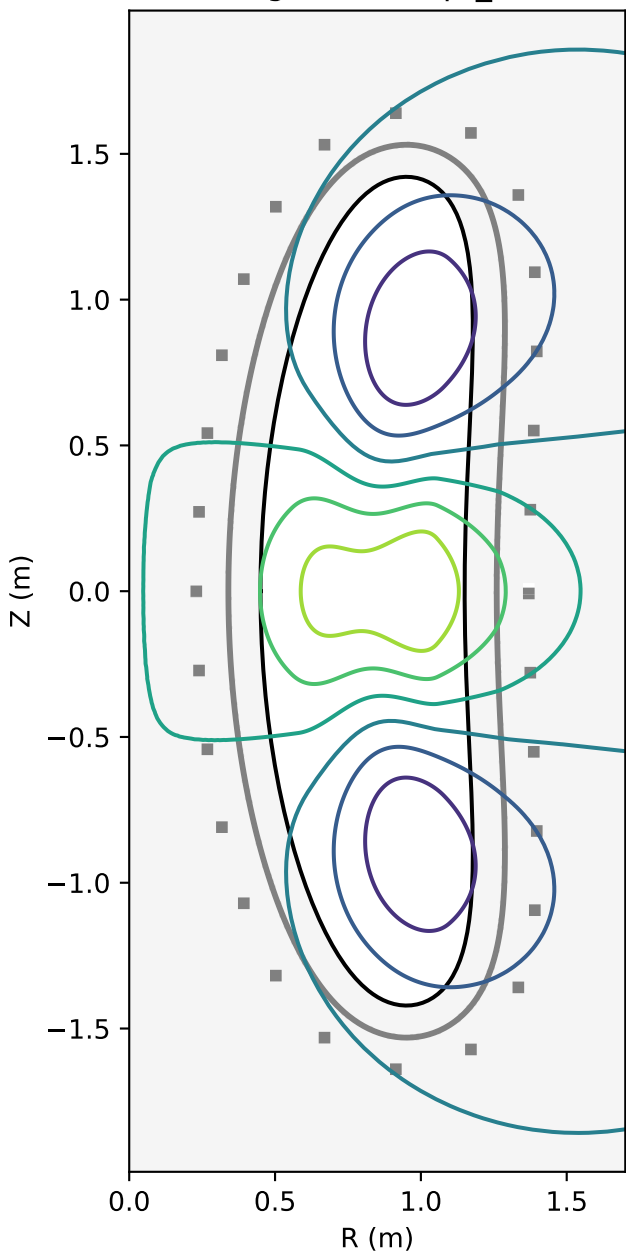
Geometry B=2.5, C=10, G=13, H=12, $\kappa=5.2$



Equilibrium flux $\kappa_{\text{actual}}=5.00$



Unstable eigenmode $\gamma\tau_w = -1.000$



Vertical Stability Analysis

Vertical Stability Results

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B=2.5 C=10 G=13 H=12

alpha_pp=2.033 gamma_pp=0.935

alpha_ff=0.938 gamma_ff=1.800

kappa_target = 5.200

kappa_actual = 5.005

beta_N = 0.415 %

growth rate (gamma) = -6.6856E+00 s-1

wall time (tau_w) = 1.4958E-01 s

gamma * tau_w = -1.0000

max coil current = 0.291 MA

Category: passive stable

axis_offset_pct = 3.64%

Equilibrium Statistics

Equilibrium Statistics:	
Topology	= Limited
Toroidal Current [A]	= 9.9998E+05
Current Centroid [m]	= 0.831 -0.000
Magnetic Axis [m]	= 0.829 0.000
Elongation	= 5.005 (U: 5.005, L: 5.005)
Triangularity	= -0.515 (U: -0.519, L: -0.512)
Plasma Volume [m^3]	= 6.244
q_0, q_95	= 1.420 13.453
Peak Pressure [Pa]	= 6.3770E+04
Stored Energy [J]	= 1.0378E+05
<Beta_pol> [%]	= 55.2885
<Beta_tor> [%]	= 0.8896
<Beta_n> [%]	= 0.4152
Diamagnetic flux [Wb]	= 7.6021E-03
Toroidal flux [Wb]	= 2.0620E+00
l_i	= 1.0904